THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET C

Chapters 13 & 14: **Statistics, Probability** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The mode of 2, 3, 3, 5, 3, 7 is: 1

(a) 2 **(b)** 7 **(c)** 5 **(d)** 3

2. In $\bar{x} = a + h\frac{\Sigma f_i u_i}{\Sigma f_i}$, h is the: 1

(a) assumed mean **(b)** class width **(c)** class mark **(d)** total frequency

3. If the mean is 30 and the mode is 24, the median is: 1

(a) 28 **(b)** 26 **(c)** 32 **(d)** 27

4. A student states: "The mean of grouped data found by the class-mark method is always exact." The correct response is: 1

(a) Incorrect — it is approximate (frequencies assumed at class marks)
(b) Exact for odd n
(c) Correct
(d) Cannot be determined

5. For $n = 50$ with cumulative frequencies 10, 22, 37, …, Student A says the median class is the 3rd class; Student B says the 2nd. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

6. The probability of an odd prime on a die is: 1

(a) $\frac{2}{3}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{1}{3}$

7. A coin is tossed three times. The probability of three heads is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{1}{8}$ **(d)** $\frac{3}{8}$

8. A letter is chosen from EDUCATION. The probability of a vowel is: 1

(a) $\frac{5}{9}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{4}{9}$

9. A student states: "Experimental and theoretical probabilities are always equal." The correct response is: 1

(a) Equal for coins
(b) Correct
(c) Incorrect — they come closer as trials increase
(d) Cannot be determined

10. For P(black king), Student A says $\frac{1}{26}$; Student B says $\frac{1}{13}$. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the mean of: 2

Class	5–15	15–25	25–35
Frequency	6	4	10

12. Find the mode of: 2

Class	0–10	10–20	20–30	30–40	40–50
Frequency	3	9	15	30	18

13. A die is thrown. Find the probability of a number strictly between 1 and 6. 2

14. In a lucky draw there are 8 prizes and 32 blanks. Find the probability of getting a blank. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the median of: 3

Class	0–10	10–20	20–30	30–40	40–50
Frequency	8	12	10	6	4

16. A card is drawn from a deck of 52. Find P(red face card), P(spade or ace) and P(not a diamond). 3

17. Find the mean by the step-deviation method: 3

Class	0–100	100–200	200–300	300–400	400–500
Frequency	4	8	12	10	6

OR

A student says the mode of the following data is 6. Verify. 3

Family size	1–3	3–5	5–7	7–9	9–11
Families	6	9	15	9	1

18. Two dice are thrown. Find P(difference 0), P(sum 6) and P(at least one 4). 3

OR

A bag has 5 white and 7 red balls. Find P(red). If one red ball is removed, find the new P(red). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** The life of 60 batteries (in hours) is given below. Find 5

(i) the mean
(ii) the median
(iii) the mode.

Life (hrs)	50–60	60–70	70–80	80–90	90–100
Batteries	6	10	18	16	10

OR

The median of the distribution is 32.5 and $\Sigma f = 40$. Find f_1 and f_2. 5

Class	0–10	10–20	20–30	30–40	40–50	50–60	60–70
Frequency	f_1	5	9	12	f_2	3	2

20. **Case study:** A game uses 20 discs numbered 1 to 20. A player wins if the number drawn is prime. 5

(i) Find P(win).
(ii) Find P(lose).
(iii) If discs numbered 21 to 25 are added, find the new P(win).
(iv) Is the game fair? Justify.

OR

A coin is tossed three times. 5

(i) Write the sample space.
(ii) Find P(exactly two heads).
(iii) Find P(at least two heads).
(iv) Find P(no tail).

END OF QUESTION PAPER — **Total: 40 Marks**